

Submission C solution to Problem 1

Abstract

We answer the question affirmatively. We build a computable countable structure $\mathcal{A}$ such that every automorphism of every copy of $\mathcal{A}$ is computable from that copy, but for every finite tuple of parameters $\bar{a}$ there is an automorphism π whose graph is not Σ_1^{in} -definable over $\bar{a} \cup \pi(\bar{a})$.

The construction

Let

$$G = [\omega]^{<\omega}$$

be the countable Boolean group of finite subsets of ω , with operation symmetric difference Δ . For $g \in G$ and $n \in \omega$, write

$$\epsilon_n(g) = \begin{cases} 1, & n \in g, \\ 0, & n \notin g. \end{cases}$$

Let $e_n = \{n\}$.

We use a computable countable language L containing:

- unary predicates $U, V_0, V_1, \dots$;
- for each $g \in G$, a binary relation symbol Q_g ;
- for each $n \in \omega$, a binary relation symbol P_n ;
- one unary function symbol S .

The domain of $\mathcal{A}$ is the disjoint union

$$U^{\mathcal{A}} \cup \bigcup_{n \in \omega} V_n^{\mathcal{A}},$$

where

$$U^{\mathcal{A}} = G$$

and

$$V_n^{\mathcal{A}} = V_n^0 \cup V_n^1, \quad V_n^i = \{v_{n,i,m} : m \in \omega\}.$$

The predicates U, V_n name these sorts. The symbols are interpreted as follows:

$$\begin{aligned} Q_g(u, v) &\iff u, v \in U^{\mathcal{A}} \text{ and } v = u \Delta g; \\ P_n(u, v_{n,i,m}) &\iff u \in U^{\mathcal{A}} \text{ and } i = \epsilon_n(u), \end{aligned}$$

and P_n is false outside $U^{\mathcal{A}} \times V_n^{\mathcal{A}}$. Finally,

$$S(u) = u \quad (u \in U^{\mathcal{A}}),$$

and

$$S(v_{n,i,m}) = v_{n,i,m+1}.$$

With any standard computable coding of G and of the triples (n, i, m) , this is a computable presentation of $\mathcal{A}$.

The automorphism group

For each $g \in G$, define a permutation τ_g of $\mathcal{A}$ by

$$\tau_g(u) = g \triangle u \quad (u \in U^{\mathcal{A}}),$$

and

$$\tau_g(v_{n,i,m}) = v_{n,i \oplus \epsilon_n(g),m}.$$

Here $\oplus$ denotes addition modulo 2.

Lemma 1. *For every $g \in G$, $\tau_g \in \text{Aut}(\mathcal{A})$, and every automorphism of $\mathcal{A}$ is of this form. Hence*

$$\text{Aut}(\mathcal{A}) = \{\tau_g : g \in G\} \cong G$$

is countable.

Proof. It is immediate from the definitions that τ_g preserves all sort predicates and preserves S .

If $Q_h(u, v)$ holds, then $v = u \triangle h$, and therefore

$$\tau_g(v) = g \triangle u \triangle h = \tau_g(u) \triangle h,$$

so $Q_h(\tau_g(u), \tau_g(v))$ holds.

Similarly, if $P_n(u, v_{n,i,m})$ holds, then $i = \epsilon_n(u)$. Since

$$\epsilon_n(g \triangle u) = \epsilon_n(g) \oplus \epsilon_n(u),$$

we have

$$P_n(\tau_g(u), \tau_g(v_{n,i,m})).$$

Thus τ_g is an automorphism.

Conversely, let $\alpha \in \text{Aut}(\mathcal{A})$. Since U is named, α maps $U^{\mathcal{A}}$ to itself. Put

$$g = \alpha(\emptyset) \in G.$$

For every $u \in G$, the relation $Q_u(\emptyset, u)$ holds. Applying α , we get

$$Q_u(g, \alpha(u)),$$

so

$$\alpha(u) = g \triangle u.$$

Now fix n and consider V_n . Let $v = v_{n,i,m}$. Choose $u \in G$ with $\epsilon_n(u) = i$. Then $P_n(u, v)$ holds. Applying α , we get

$$P_n(g \triangle u, \alpha(v)).$$

Therefore $\alpha(v)$ lies in the copy $V_n^{i \oplus \epsilon_n(g)}$.

Finally, α preserves S . On each copy V_n^i , the map S is the successor map on a one-way chain of type ω . Any bijection between two such chains preserving successor must send the unique element not in the range of S to the unique element not in the range of S , and hence must preserve the coordinate m . Thus

$$\alpha(v_{n,i,m}) = v_{n,i \oplus \epsilon_n(g),m}.$$

So $\alpha = \tau_g$. □

Every automorphism of every copy is relatively computable

Proposition 1. *For every copy $\mathcal{B} \cong \mathcal{A}$ and every $\beta \in \text{Aut}(\mathcal{B})$, the function β is computable relative to the atomic diagram of $\mathcal{B}$. In particular, $\mathcal{A}$ is computably Aut-countable on the cone with base $\emptyset$.*

Proof. Let $\mathcal{B}$ be any copy of $\mathcal{A}$, and let $\beta \in \text{Aut}(\mathcal{B})$. By the previous lemma, transferred to $\mathcal{B}$, there is a unique $g \in G$ such that β acts on the U -sort as translation by g . Since g is a finite subset of ω , it can be hard-coded into a Turing program computing β .

Choose any $u_0 \in U^{\mathcal{B}}$ and let

$$u_1 = \beta(u_0).$$

The program may hard-code the natural numbers u_0, u_1 .

Given input $x \in U^{\mathcal{B}}$, search through $h \in G$ until finding the unique h such that

$$\mathcal{B} \models Q_h(u_0, x).$$

Then search for the unique y such that

$$\mathcal{B} \models Q_h(u_1, y).$$

Output this y . This computes $\beta(x)$ for $x \in U^{\mathcal{B}}$.

Now suppose $x \in V_n^{\mathcal{B}}$. We can find n by searching the unary predicates $V_0, V_1, \dots$.

If $n \notin g$, then β is the identity on $V_n^{\mathcal{B}}$, so output x .

If $n \in g$, then β swaps the two S -chains inside $V_n^{\mathcal{B}}$. Choose some

$$p_n \in V_n^{\mathcal{B}}$$

and let

$$q_n = \beta(p_n).$$

The program may hard-code the finitely many pairs (p_n, q_n) for $n \in g$.

To compute $\beta(x)$ for $x \in V_n^{\mathcal{B}}$ with $n \in g$, dovetail the following searches:

- search for d with $S^d(p_n) = x$; if found, output $S^d(q_n)$;
- search for d with $S^d(x) = p_n$; if found, search for the unique y with $S^d(y) = q_n$, and output y ;
- search for d with $S^d(q_n) = x$; if found, output $S^d(p_n)$;
- search for d with $S^d(x) = q_n$; if found, search for the unique y with $S^d(y) = p_n$, and output y .

Exactly one of these alternatives eventually applies, because each V_n consists of two one-way S -chains and p_n, q_n are corresponding points in the two chains.

All searches use only the atomic diagram of $\mathcal{B}$. Thus β is computable relative to $\mathcal{B}$. □

A preservation lemma for existential formulas

Fix n . For $d_0, d_1 \in \omega$, define a map

$$\rho_{d_0, d_1}^n : \mathcal{A} \rightarrow \mathcal{A}$$

by fixing every element outside V_n , and by

$$\rho_{d_0, d_1}^n(v_{n, i, m}) = v_{n, i, m + d_i}.$$

Lemma 2. *For every n and every $d_0, d_1 \in \omega$, the map ρ_{d_0, d_1}^n is an embedding of $\mathcal{A}$ into itself. Consequently, if an existential formula with parameters outside V_n holds of a tuple, then it still holds after applying ρ_{d_0, d_1}^n to the non-parameter entries.*

Proof. The map ρ_{d_0, d_1}^n is injective. It clearly preserves the unary predicates. It fixes U , so it preserves every Q_g . It also preserves S , because

$$\rho_{d_0, d_1}^n(S(v_{n, i, m})) = v_{n, i, m + 1 + d_i} = S(\rho_{d_0, d_1}^n(v_{n, i, m})).$$

Finally, $P_n(u, v_{n, i, m})$ depends only on u and on the copy index i , and ρ_{d_0, d_1}^n fixes u and preserves i . For $m \neq n$, P_m is unaffected.

Thus ρ_{d_0, d_1}^n is an L -embedding. Embeddings preserve quantifier-free formulas, and therefore preserve existential formulas. $\square$

Failure of Σ_1^{in} -definability

Lemma 3. *Let $\bar{p}$ be a finite tuple from $\mathcal{A}$ containing no element of V_n . Then the automorphism τ_{e_n} is not Σ_1^{in} -definable over $\bar{p}$.*

Proof. Let

$$r_0 = v_{n, 0, 0}, \quad r_1 = v_{n, 1, 0}.$$

Then

$$\tau_{e_n}(r_0) = r_1.$$

Suppose, toward a contradiction, that the graph of τ_{e_n} is Σ_1^{in} -definable over $\bar{p}$. Then there is a countable family of existential formulas

$$(\varphi_i(\bar{p}, x, y))_{i \in \omega}$$

such that

$$\tau_{e_n}(x) = y \iff \mathcal{A} \models \varphi_i(\bar{p}, x, y) \text{ for some } i.$$

Since $\tau_{e_n}(r_0) = r_1$, there is some i with

$$\mathcal{A} \models \varphi_i(\bar{p}, r_0, r_1).$$

Now apply the embedding $\rho_{0, 1}^n$. It fixes $\bar{p}$ and r_0 , and sends

$$r_1 \mapsto S(r_1) = v_{n, 1, 1}.$$

By preservation of existential formulas under embeddings,

$$\mathcal{A} \models \varphi_i(\bar{p}, r_0, S(r_1)).$$

But

$$\tau_{e_n}(r_0) = r_1 \neq S(r_1),$$

so $(r_0, S(r_1))$ is not in the graph of τ_{e_n} . This contradicts the assumed definition of the graph. $\square$

Main theorem

Theorem 1. *There is a countable computably represented structure $\mathcal{A}$ such that:*

1. *$\mathcal{A}$ is computably Aut-countable on a cone;*
2. *for every finite tuple of parameters $\bar{a}$ from $\mathcal{A}$, there is an automorphism $\pi \in \text{Aut}(\mathcal{A})$ such that π is not Σ_1^{in} -definable over $\bar{a} \cup \pi(\bar{a})$.*

Proof. The structure $\mathcal{A}$ constructed above is computable and countable. By the proposition, every automorphism of every copy of $\mathcal{A}$ is computable relative to that copy, so $\mathcal{A}$ is computably Aut-countable on the cone with base $\emptyset$.

Now let $\bar{a}$ be any finite tuple from $\mathcal{A}$. Since $\bar{a}$ is finite, it meets only finitely many of the sorts V_n . Choose n such that no entry of $\bar{a}$ lies in V_n .

Let

$$\pi = \tau_{e_n}.$$

Then π toggles the two copies inside V_n and fixes every V_m with $m \neq n$. Since $\bar{a}$ has no entry in V_n , the tuple $\pi(\bar{a})$ also has no entry in V_n . Therefore the finite tuple

$$\bar{p} = \bar{a} \cup \pi(\bar{a})$$

contains no element of V_n .

By the previous lemma, the graph of $\pi = \tau_{e_n}$ is not Σ_1^{in} -definable over $\bar{p}$. Hence π is not Σ_1^{in} -definable over $\bar{a} \cup \pi(\bar{a})$.

Thus $\mathcal{A}$ has the required properties. □